HugAgent

A Human Simulation Benchmark for Individual-Level Reasoning

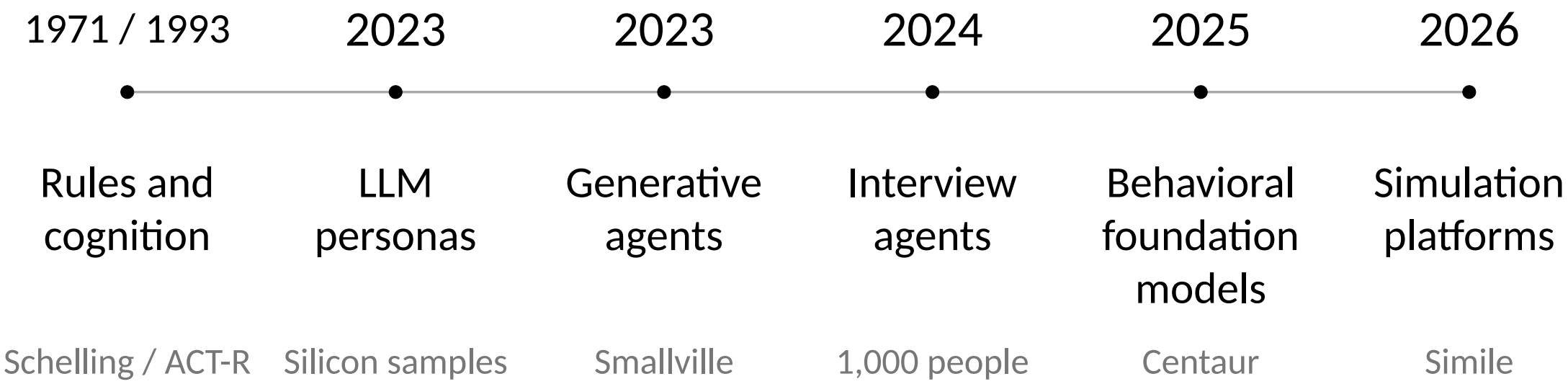
EMNLP 2026 · Oral presentation

Chance Jiajie Li*, Zhenze Mo*, Yuhan Tang*, Ao Qu, Jiayi Wu, Kaiya Ivy Zhao, Yulu Gan,
Jie Fan, Jiangbo Yu, Hang Jiang, Paul Pu Liang, Jinhua Zhao,
Luis Alberto Alonso Pastor, Kent Larson

* Equal contribution

Human Simulation and Digital Twins

- A testbed for policy, products and social research.

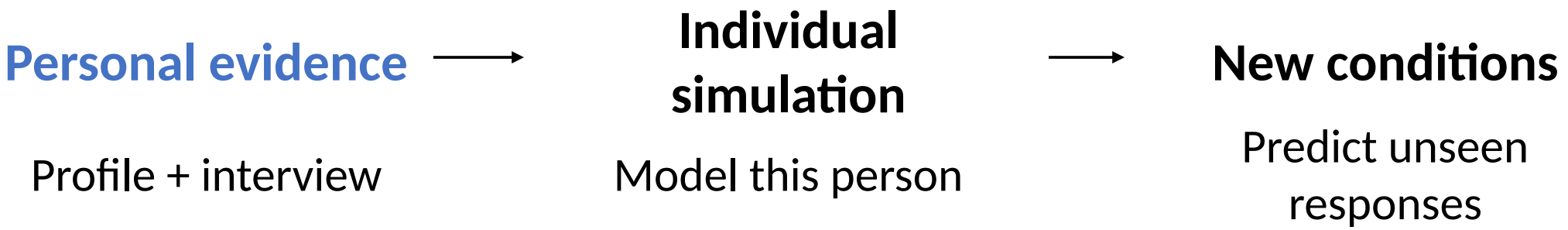


How well does an agent represent this person?

Selected milestones; schematic timeline. Schelling (1971); Anderson (1993); Argyle & Park (2023); Park (2024); Binz (2025); Simile / Gallup (2026).

Can Agents Represent an Individual?

- LLMs simulate participants in social research.
- Interviews ground agents in **real individuals**.



Can an agent predict this person's responses when **conditions change**?

HugAgent evaluates individual beliefs and their updates.

Same Answers, Different Reasons

- Should the city expand public surveillance?



Person A

3 / 10

Privacy

Data retention



Person B

3 / 10

Trust

Institutional reliability

Matching stance scores does not establish reasoning fidelity.

Hypothetical people and 0–10 support ratings, not participant data. Portraits are AI-generated illustrations.

How Will Each Person Respond?

- Same people and reasons. Change one condition.



Person A

3 → ?

Privacy

Data retention



Person B

3 → ?

Trust

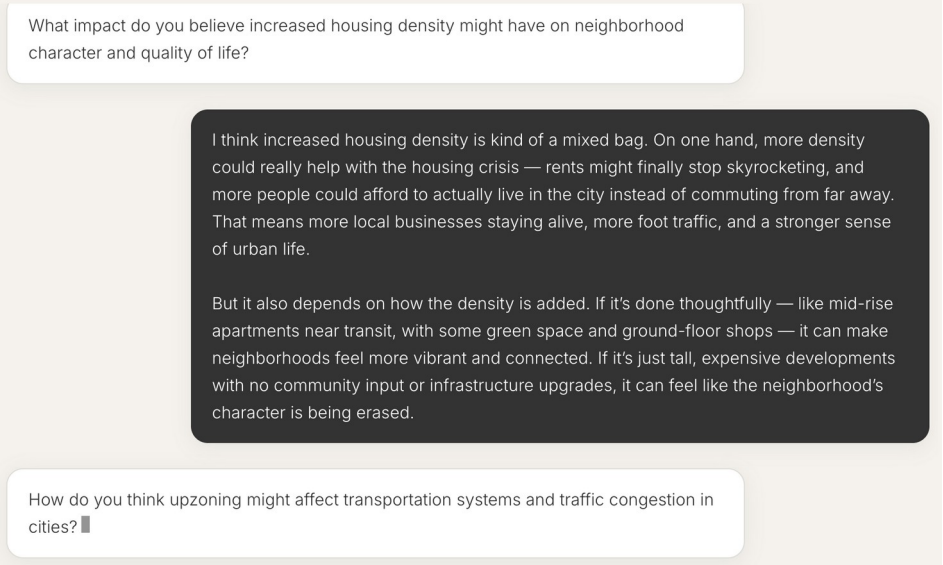
Institutional reliability

New condition: **delete footage after 48 hours**

Constructed example. Question marks are responses to predict, not observed outcomes. Portraits are AI-generated.

Human Responses as Ground Truth

- Baseline survey
  Current beliefs and reasons
- Hypothetical interventions
  Held-out updated responses
- Adaptive interview
  Personal reasoning transcript
- Human retest
  Empirical consistency reference



Adaptive Interview Tool

54	3	1,742
screened participants	policy domains	evaluation items

Healthcare, surveillance and zoning. Retest after ~14 days: 18 completed, 13 retained. Interview interface: HugAgent paper.

Task 1: Inferring Existing Beliefs

- What do A and B already believe?



Person A

Current support

3 / 10



Person B

Current support

3 / 10

Data retention → + / – ? → Support

Institutional reliability → + / – ? → Support

Infer whether each factor raises or lowers support

356 items

Inputs: demographics, interview and permitted non-target survey. Targets withheld. Constructed example with AI portraits.

Task 2: Predicting Belief Updates

- New condition: **delete footage after 48 hours**



Person A

New support

3 → ?

Data retention

Importance: ?



Person B

New support

3 → ?

Institutional reliability

Importance: ?

Predict updated stance and reason importance

1,386 items

Same constructed example, not observed outcomes. Predict updated stance and reason importance; target answers are withheld.

The Person's Own Context Helps

- Does knowing the correct person help?

Context	Update accuracy (%)
Demographics only	39.83 ±0.45
Another person's interview	39.30 ±0.22
Own context	**63.11 ±0.14**

- Own context: **+23.28 percentage points**
- The cross-person control weakens a "more text" explanation.

GPT-4o belief-update accuracy (%), mean ± SD over 5 runs. Own context includes permitted non-target survey answers.

Belief Updates Leave a Larger Gap

- How closely do models match the same person?

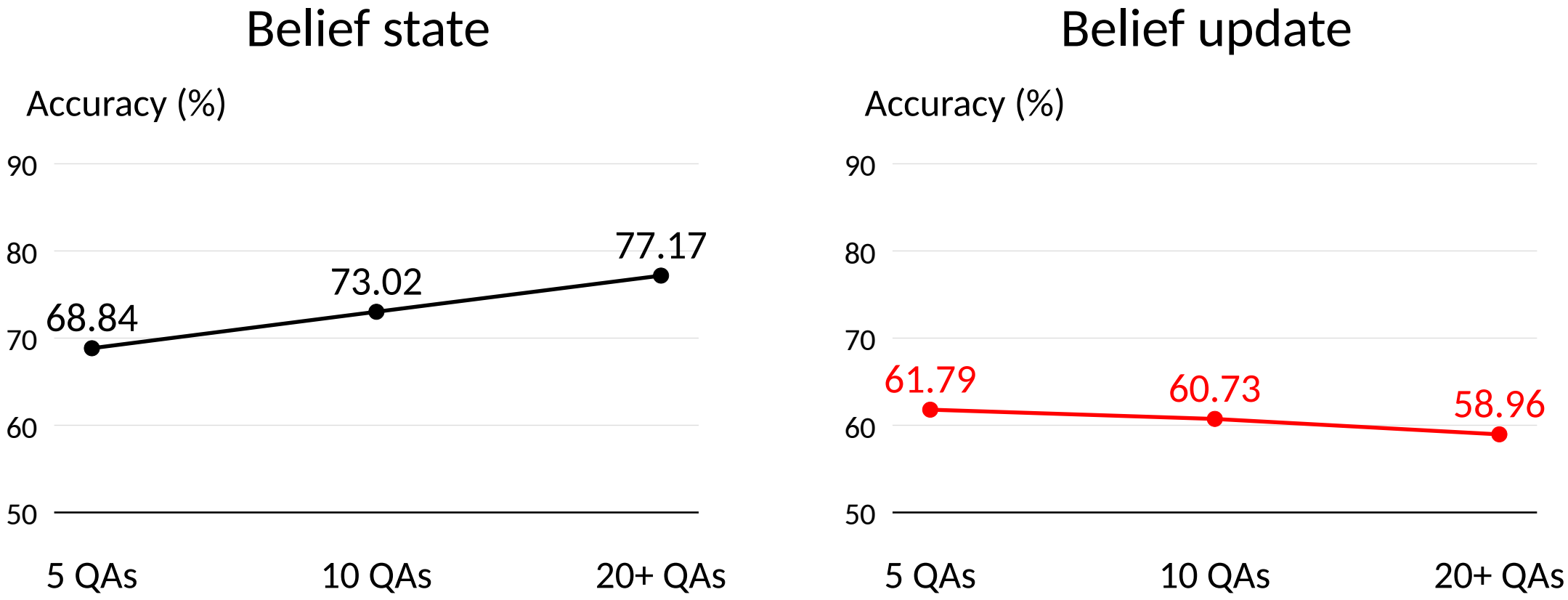
Accuracy (%)

Task	GPT-4o	Human retest	Gap (pp)
Belief state	74.66	84.84	10.18
Belief update	**63.11**	**85.66**	**22.55**

- Belief updates leave a **larger gap**.
- Compare each model score with its own task's human reference.

State: factor-polarity accuracy. Update: tolerance-band accuracy (±2 stance; ±1 reason). Human retest after ~14 days.

Does More Context Close the Gap?



- More context helps states, but **not updates**.

Qwen2.5-32B. Context QAs supplied to the model; this does not vary original interview duration. Means over 5 runs.

Diagnosing Belief Updates

GPT-4o on healthcare updates

Will the answer change?

Person: 3 → 6 Model: 3 → 3

Missed change

Change-detection accuracy

49.36%

Which direction?

Person: 3 → 6 Model: 3 → 4

Correct upward direction

Direction-inference accuracy

88.89%

Hypothesis: models struggle to predict when a person changes.

Mean over 5 runs. Direction accuracy is conditional on both predicted and reported change. Examples are illustrative, not participant data.

Reliable Individual Simulation

- **Personal context** supports individual simulation.

  Own context improves update accuracy by 23.28 pp.

- **New conditions** expose a remaining gap.

  Update accuracy remains 22.55 pp below the human retest reference.

- **More transcript** does not reliably close it.

  Which personal information helps predict belief changes?

Scope: self-reported hypothetical responses; 54 people, 3 policy domains.

HugAgent

Interviews ground simulation.

Counterfactuals test its limits.

Open source: benchmark, data, evaluation code and Adaptive Interview Tool

Paper

Project & code